



syed123@localhost:~ — nano pc.c



GNU nano 5.6.1

pc.c

Modified

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

int buffer[N], in = 0, out = 0;
sem_t empty, full;
pthread_mutex_t m;

void *prod() {
    for(int i=0;i<10;i++) {
        sem_wait(&empty);
        pthread_mutex_lock(&m);

        buffer[in] = i;
        printf("Produced: %d\n", i);
        in = (in+1)%N;
```



GNU nano 5.6.1

receiver.c

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>

int main() {
    int shmid;
    char *str;

    // Access shared memory
    shmid = shmget((key_t)1234, 1024, 0666);

    // Attach
    str = (char*) shmat(shmid, NULL, 0);

    // Read message
    printf("Message Received: %s\n", str);

    // Detach
    shmdt(str);
}
```



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pc.c

Modified

```
    out = (out+1)%N;

    pthread_mutex_unlock(&m);
    sem_post(&empty);
    sleep(2);
}

int main() {
    pthread_t p,c;

    sem_init(&empty,0,N);
    sem_init(&full,0,0);
    pthread_mutex_init(&m,NULL);

    pthread_create(&p,NULL,prod,NULL);
    pthread_create(&c,NULL,cons,NULL);

    pthread_join(p,NULL);
    pthread_join(c,NULL);
}
```

```
[syed123@localhost ~]$ ./pc
```

1. Producer
2. Consumer
3. Exit

Enter your choice: 2

Buffer is empty!!

1. Producer
2. Consumer
3. Exit

Enter your choice: 3

```
[syed123@localhost ~]$
```